SIMATS ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

MD Imthiaz I
192521360RD

CO2-AT1-Assignment1

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QUESTIONS

Q1
Smart Document Editor using String, StringBuilder and StringBuffer
10 marks

Q2
Hospital Token Management System using List Interface
10 marks

Q3
Airport Boarding Queue Management using Queue Interface
10 marks

Q4
Digital Certificate Verification using Set Interface
10 marks

Q5
Smart Inventory Management using Map Interface
10 marks

5/5 answered

Q1 **Smart Document Editor using String, StringBuilder and StringBuffer** 10 marks

Question 1 :
1.Read a paragraph from the user.
2.Count the number of characters, words, and sentences.
3.Replace every occurrence of a given word with another word.
4.Reverse each sentence individually.
5.Maintain editing history using StringBuffer.
6.Construct the final formatted document using StringBuilder.
7.Display the original document, editing history, and final document.
Requirement: Use String methods wherever appropriate. Use StringBuilder for document construction and StringBuffer for edit history.

Your Code

```
import java.util.Scanner;
public class DocumentEditor {
    public static void main(String[] args)
        Scanner sc = new Scanner(System.ir
        System.out.println("Enter a paragı
        String paragraph = sc.nextLine();
        int characters = paragraph.length
        String words[] = paragraph.trim().
```

```
        int wordCount = words.length;
        String sentences[] = paragraph.sp]
        int sentenceCount = 0;
        for (String s : sentences) {
            if (!s.trim().isEmpty())
                sentenceCount++;
        }
        System.out.println("\nCharacters :
        System.out.println("Words      : '
        System.out.println("Sentences  : '
        System.out.print("\nEnter word to
        String oldWord = sc.next();
        System.out.print("Enter new word:
        String newWord = sc.next();
        String replacedParagraph = paragra
        StringBuffer history = new StringI
        history.append("Original Paragrapl
        history.append(paragraph);
        history.append("\n\nReplaced '");
        history.append(oldWord);
        history.append("' with '");
        history.append(newWord);
        history.append("'\n");
        history.append(replacedParagraph);
        StringBuilder finalDocument = new
        String reverseSentences[] = replac

        for (String sentence : reverseSent
            String temp = sentence.trim();
            if (!temp.isEmpty()) {
                StringBuilder rev = new St
                finalDocument.append(rev.1
            }
        }
        System.out.println("\n========== (
        System.out.println(paragraph);
        System.out.println("\n========== I
        System.out.println(history);
        System.out.println("\n========== I
        System.out.println(finalDocument.t
        sc.close();
    }
}
```

Input

```
Java is easy to learn. Java is object oriented.
It is widely used!
Java
Python
```

Output

```
Enter a paragraph:

Characters : 66
Words      : 13
Sentences  : 3
```

Run Save

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5/5 answered

Q2 **Hospital Token Management System using List Interface** 10 marks

Hospital Token Management System using List Interface

Store patient details using an appropriate List implementation.

Register new patients.

Display patients in registration order.

Update patient information using Patient ID.

Remove cancelled appointments.

Display all patients of a particular department.

Find the first and last registered patients.

Requirement: Use a suitable List implementation

Your Code

```
import java.util.ArrayList;
import java.util.List;
import java.util.Scanner;
public class HospitalTokenManagement {
    public static void main(String[] args)
        Scanner sc = new Scanner(System.in
        List<Patient> patients = new Array
        System.out.print("Enter number of
        int n = sc.nextInt();
```

```
        sc.nextLine();
        for (int i = 0; i < n; i++) {
            System.out.println("\nEnter Pa
            System.out.print("Patient ID:
            int id = sc.nextInt();
            sc.nextLine();
            System.out.print("Name: ");
            String name = sc.nextLine();
            System.out.print("Department:
            String dept = sc.nextLine();
            patients.add(new Patient(id, n
        }
        System.out.println("\n--- Patients
        for (Patient p : patients) {
            p.display();
        }
        System.out.print("Enter Patient II
        int updateId = sc.nextInt();
        sc.nextLine();
        boolean found = false;
        for (Patient p : patients) {
            if (p.id == updateId) {
                System.out.print("Enter Ne
                p.name = sc.nextLine();
                System.out.print("Enter Ne
                p.department = sc.nextLine
                found = true;
                System.out.println("Patien
                break;
            }
        }
        if (!found) {
            System.out.println("Patient II
        }
        System.out.print("\nEnter Patient
        int removeId = sc.nextInt();
        found = false;
        for (int i = 0; i < patients.size
            if (patients.get(i).id == remo
                patients.remove(i);
                found = true;
                System.out.println("Appoin
                break;
            }
        }
        if (!found) {
            System.out.println("Patient II
        }
```

```
        sc.nextLine();
        System.out.print("\nEnter Departme
        String searchDept = sc.nextLine();
        System.out.println("\nPatients in
        found = false;
        for (Patient p : patients) {
            if (p.department.equalsIgnore(
                p.display();
                found = true;
            }
        }
        if (!found) {
            System.out.println("No patient
        }
        if (!patients.isEmpty()) {
            System.out.println("--- First
            patients.get(0).display();
            System.out.println("--- Last I
            patients.get(patients.size() -
        }
        sc.close();
    }
}
class Patient {
    int id;
    String name;
    String department;

    Patient(int id, String name, String de
        this.id = id;
        this.name = name;
        this.department = department;
    }
    void display() {
        System.out.println("Patient ID : '
        System.out.println("Name       : '
        System.out.println("Department : '
        System.out.println();
    }
}
```

Input

```
1
101
Arun
Cardiology
```

Output

```
Enter number of patients:
Enter Patient 1 Details
Patient ID: Name: Department:
--- Patients in Registration Order ---
Patient ID : 101
Name       : Arun
```

Run Save

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QUESTIONS

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Board passengers in FIFO order.

Display the next passenger without removing them.

Handle VIP passengers using a priority-based queue.

Display remaining passengers after each boarding operation.

Requirement: Use appropriate Queue implementations.

Q3
Airport Boarding Queue Management using Queue Interface
10 marks

Q4
Digital Certificate Verification using Set Interface
10 marks

Q5
Smart Inventory Management using Map Interface
10 marks

5/5 answered

Your Code

```
import java.util.LinkedList;
import java.util.PriorityQueue;
import java.util.Queue;
import java.util.Scanner;
public class AirportBoardingQueue {
    public static void main(String[] args)
        Scanner sc = new Scanner(System.i
        Queue<String> boardingQueue = new
        PriorityQueue<String> vipQueue =
        System.out.print("Enter number of
```

```
        int n = sc.nextInt();
        sc.nextLine();
        for (int i = 0; i < n; i++) {
            System.out.print("Enter Passe
            boardingQueue.offer(sc.nextLi
        }
        System.out.print("\nEnter number c
        int m = sc.nextInt();
        sc.nextLine();
        for (int i = 0; i < m; i++) {
            System.out.print("Enter VIP Pa
            vipQueue.offer(sc.nextLine());
        }
        System.out.println("\n--- Normal I
        System.out.println(boardingQueue);
        System.out.println("\n--- VIP Queu
        System.out.println(vipQueue);
        if (!vipQueue.isEmpty()) {
            System.out.println("\nNext VII
        }
        if (!boardingQueue.isEmpty()) {
            System.out.println("Next Norma
        }
        System.out.println("\n--- Boardin
        while (!vipQueue.isEmpty() || !boa
            if (!vipQueue.isEmpty()) {
                System.out.println("Boarde
            } else {
                System.out.println("Boarde
            }
            System.out.println("Remaining
            System.out.println("Remaining
            System.out.println();
        }
        System.out.println("All passengers
        sc.close();
    }
}
```

Input

```
3
Arun
Priya
Rahul
```

Output

```
[Arun, Priya, Rahul]

--- VIP Queue ---
[Aman, Zoya]

Next VIP Passenger: Aman
```

Run Save

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QUESTIONS

Q1
Smart Document Editor using String, StringBuilder and StringBuffer
10 marks

Q4 **Digital Certificate Verification using Set Interface** 10 marks

Digital Certificate Verification using Set Interface

Store certificate IDs using an appropriate Set implementation

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Q4
Digital Certificate Verification using Set Interface
10 marks

Requirement: Use suitable Set implementations.

Q5
Smart Inventory Management using Map Interface
10 marks

5/5 answered

Your Code

```
import java.util.*;
public class DigitalCertificateVerificatic
    public static void main(String[] args)
        Scanner sc = new Scanner(System.ir
        LinkedHashSet<String> certificate:
        System.out.print("Enter the numbe:
        int n = sc.nextInt();
        System.out.println("Enter Certific
        for (int i = 0; i < n; i++) {
            certificates.add(sc.next());
        }
```

```
        System.out.println("\nCertificates
        for (String cert : certificates)
            System.out.print(cert + " ");
        }
        System.out.print("\n\nEnter Certi
        String search = sc.next();
        if (certificates.contains(search))
            System.out.println("Certificat
        } else {
            System.out.println("Certificat
        }
        TreeSet<String> sorted = new TreeS
        System.out.println("\nCertificates
        for (String cert : sorted) {
            System.out.print(cert + " ");
        }
        System.out.println("\n\nTotal Uni
        System.out.print("\nEnter Revoked
        String revoked = sc.next();
        if (certificates.remove(revoked))
            System.out.println("Certificat
        } else {
            System.out.println("Certificat
        }
        System.out.println("\nRemaining Ce
        for (String cert : certificates)
            System.out.print(cert + " ");
        }
        sc.close();
    }
}
```

Input

```
CERT104
CERT102
CERT103
CERT102
```

Output

```
Enter the number of certificate IDs: Enter
Certificate IDs:

Certificates in Insertion Order:
CERT101 CERT102 CERT103 CERT104
```

Run Save

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Q5
Smart Inventory Management using Map Interface
10 marks

5/5 answered

Q5 **Smart Inventory Management using Map Interface** 10 marks

Smart Inventory Management using Map Interface

Store product information using an appropriate Map implementation.

Add new products.

Update product quantity.

Search a product using Product ID.

Calculate the total inventory value.

Display products category-wise.

Remove discontinued products.

Display all available products.

Requirement: Use a suitable Map implementation.

Your Code

```
import java.util.*;
public class SmartInventoryManagement {
    public static void main(String[] args)
        Scanner sc = new Scanner(System.i
        HashMap<Integer, Product> invento
        System.out.print("Enter number of
        int n = sc.nextInt();
        for (int i = 0; i < n; i++) {
            System.out.println("\nEnter P
```

```
            int id = sc.nextInt();
            sc.nextLine();
            System.out.print("Enter Produ
            String name = sc.nextLine();
            System.out.print("Enter Categ
            String category = sc.nextLine
            System.out.print("Enter Quant
            int quantity = sc.nextInt();
            System.out.print("Enter Price
            double price = sc.nextDouble(
            inventory.put(id, new Product
        }
        System.out.print("\nEnter Product
        int updateId = sc.nextInt();
        if (inventory.containsKey(updateI
            System.out.print("Enter New Q
            inventory.get(updateId).quant
            System.out.println("Quantity
        } else {
            System.out.println("Product N
        }
        System.out.print("\nEnter Product
        int searchId = sc.nextInt();
        if (inventory.containsKey(searchI
            Product p = inventory.get(sea
            System.out.println("Product F
            System.out.println(p.name + "
        } else {
            System.out.println("Product N
        }
        double total = 0;
        for (Product p : inventory.values
            total += p.quantity * p.price;
        }
        System.out.println("\nTotal Inven
        System.out.println("\nProducts Ca
        LinkedHashMap<String, ArrayList<St
        for (Map.Entry<Integer, Product>
            Product p = entry.getValue();
            if (!categoryMap.containsKey(
                categoryMap.put(p.categor
            }
            categoryMap.get(p.category).a
        }
        for (String cat : categoryMap.key
            System.out.println(cat + " :
        }
        System.out.print("\nEnter Product
```

```
        int removeId = sc.nextInt();
        if (inventory.remove(removeId) !=
            System.out.println("Product Re
        } else {
            System.out.println("Product No
        }
        System.out.println("\nAvailable Pr
        for (Map.Entry<Integer, Product> e
            Product p = entry.getValue();
            System.out.println(entry.getKe
                    " Qty:" + p.quantity -
        }
        sc.close();
    }
}
class Product {
    String name;
    String category;
    int quantity;
    double price;
    Product(String name, String category,
        this.name = name;
        this.category = category;
        this.quantity = quantity;
        this.price = price;
    }
}

```

Input

```
2000
102
15
101
103
```

Output

```
Enter number of products:
Enter Product ID:
Enter Product Name: Enter Category: Enter
Quantity: Enter Price:
Enter Product ID:
Enter Product Name: Enter Category: Enter
```

Run Save